(i) Using Heron's formula , Pythagorean theorem can be proved for right triangle and conversely, using Pythagorean theorem for right triangle one can establish Heron's area formula.

(ii) If area of a triangle is given as an integer, then Heron's formula is useful in finding triangles with integer sides.

(iii) If the perimeter of a triangle is fixed, then Heron's formula is useful for finding triangles having integer area and integer sides.

For example, if the perimeter of a triangle is $100m$, then there is a triangle with sides $32\ m, 34\ m, 34\ m$ and area $480\ m^2$.

Example 3.57 In a $\triangle ABC$, prove that $b^2 \sin 2C + c^2 \sin 2B = 2bc \sin A$.

Solution:

The Sine formula is, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$

Thus, $a = 2R \sin A; \quad b = 2R \sin B; \quad c = 2R \sin C$

$$\begin{aligned} b^2 \sin 2C + c^2 \sin 2B &= 4R^2 \sin^2 B \sin 2C + 4R^2 \sin^2 C \sin 2B \\ &= 4R^2 \left(2 \sin^2 B \sin C \cos C + 2 \sin^2 C \sin B \cos B\right) \\ &= 8R^2 \sin B \sin C \left(\sin B \cos C + \sin C \cos B\right) \\ &= 8R^2 \sin B \sin C \sin(B + C) \\ &= 8R^2 \sin B \sin C \sin(\pi - A) = 8R^2 \sin B \sin C \sin A \\ &= 8R^2 \left(\frac{b}{2R}\right)\left(\frac{c}{2R}\right) \sin A = 2bc \sin A. \end{aligned}$$

Example 3.58 In a $\triangle ABC$, prove that $\sin\left(\dfrac{B-C}{2}\right) = \dfrac{b-c}{a} \cos\dfrac{A}{2}$.

Solution:

The Sine formula is, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$

Now, $$\begin{aligned} \frac{b-c}{a} \cos\frac{A}{2} &= \frac{2R \sin B - 2R \sin C}{2R \sin A} \cos\frac{A}{2} \\ &= \frac{2 \sin\left(\dfrac{B-C}{2}\right) \cos\left(\dfrac{B+C}{2}\right)}{2 \sin\dfrac{A}{2} \cos\dfrac{A}{2}} \cos\frac{A}{2} \\ &= \frac{\sin\left(\dfrac{B-C}{2}\right) \cos\left(90^\circ - \dfrac{A}{2}\right)}{\sin\dfrac{A}{2}} \\ &= \frac{\sin\left(\dfrac{B-C}{2}\right) \sin\left(\dfrac{A}{2}\right)}{\sin\dfrac{A}{2}} \\ &= \sin\left(\frac{B-C}{2}\right). \end{aligned}$$